

Shape-First Training-Rule Distillation Pilot Campaign Results

Shape-First Training-Rule Distillation Pilot Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

The `shape_first_training_rule_distillation_pilot_2026_07_22` campaign tested whether stable checks from the `TE Curve Verification Pipeline` shape-first screen could be used as auxiliary training terms instead of only post-training acceptance criteria.

The pilot deliberately kept both modeling roads alive:

- one time-windowed `periodic_gru_sequence` candidate;
- one non-windowed `periodic_mlp` harmonic candidate.

The campaign ran remotely on `xilab-remote` against `polished_dataset`, setpoint inputs, and forward-only `Fw` curves. Both runs completed successfully. The local terminal remained active after remote completion because the shared remote launcher used a blocking SSH stream polling path. That infrastructure issue was fixed in `scripts/campaigns/infrastructure/run_remote_training_campaign.ps1` by moving remote `stdout/stderr` handling to asynchronous stream events.

Execution Summary

Item	Value
Campaign	shape_first_training_rule_distillation_pilot_2026_07_22
Remote host	xilab-remote
Dataset	polished_dataset
Input mode	setpoints
Surface	Fw
Started	2026-07-22T14:38:51+02:00
Completed	2026-07-22T14:48:11+02:00
Completed runs	2
Failed runs	0
Campaign output	output/training_campaigns/2026-07-22-14-38-51_shape_first_training_rule_distillation_pilot_2026_07_22

Candidate Ranking

Rank	Family	Model Type	Windowed	Validation MAE	Test MAE	Test RMSE	Params
1	shape_first_distilled_periodic_mlp_harmonic_fw	periodic_mlp	no	0.001573	0.001420	0.001866	28,545
2	shape_first_distilled_periodic_gru_sequence_fw	periodic_gru_sequence	yes	0.002004	0.001523	0.001920	157,953

Scalar campaign selection therefore picks the non-windowed harmonic MLP. This is not a promotion decision: the repository selection policy requires curve-first evidence before accepting a compensation-relevant forward model.

Metric Breakdown

Metric	Time-Windowed GRU	Non-Windowed Harmonic MLP
Validation pointwise loss	0.017261	0.009553
Validation centered shape loss	0.014137	0.007614
Validation offset loss	0.003124	0.003152
Validation amplitude loss	0.046970	0.102087
Validation sparse harmonic loss	0.000339	0.000131
Test pointwise loss	0.009925	0.007084
Test centered shape loss	0.007235	0.005375
Test offset loss	0.002689	0.003210
Test amplitude loss	0.018799	0.075478
Test sparse harmonic loss	0.000150	0.000090

The non-windowed harmonic MLP improved scalar MAE, pointwise loss, mean-centered shape loss, and sparse harmonic loss. The time-windowed GRU kept better test offset and amplitude losses. That split is exactly why the result needs a bounded `TE Curve Verification Pipeline` screen before any expansion: the scalar and shape-component signals do not collapse to a single clear deployment answer.

Interpretation

This pilot gives useful evidence for the training-rule distillation idea. The non-windowed harmonic road was not dead: once the shape-first rules were added as training terms, it became the scalar winner and slightly beat the recent shape-gate loss v2 checkpoint-selection pilot on test MAE.

The result does not overturn the current forward recommendation. A prior non-windowed shape-objective scalar winner failed the bounded curve-first screen, so scalar improvement alone is not enough. The correct reading is: continue testing both the time-windowed and non-windowed roads, but force the new scalar winner through the same bounded curve-first gate before any larger campaign.

Decision

The campaign is accepted as a completed pilot. The scalar winner is

`shape_first_distilled_periodic_mlp_harmonic_fw`.

Do not promote it yet. The next required step is a bounded `TE Curve Verification Pipeline` screen on `polished_dataset`, setpoint inputs, and `Fw`, comparing:

- `polished_setpoints_periodic_gru_sequence_fw` ;
- `polished_setpoints_periodic_mlp_harmonic_fw` ;
- `shape_first_distilled_periodic_gru_sequence_fw` ;
- `shape_first_distilled_periodic_mlp_harmonic_fw` .

That screen should report raw error, centered shape fidelity, offset behavior, harmonic retention, derivative/phase behavior, visual measured-vs-predicted curves, and the curve-first recommendation.

Artifacts

- Campaign execution report:
`output/training_campaigns/2026-07-22-14-38-51_shape_first_training_rule_distillation_pilot_2026_07_22/campaign_execution_report.md`
- Campaign leaderboard:
`output/training_campaigns/2026-07-22-14-38-51_shape_first_training_rule_distillation_pilot_2026_07_22/campaign_leaderboard.yaml`
- Campaign best run:
`output/training_campaigns/2026-07-22-14-38-51_shape_first_training_rule_distillation_pilot_2026_07_22/campaign_best_run.yaml`
- Time-windowed GRU run:
`output/training_runs/shape_first_training_rule_distillation/2026-07-22-14-38-51_te_shape_first_distilled_periodic_gru_sequence_fw_polished_setpoints`
- Non-windowed harmonic MLP run:
`output/training_runs/shape_first_training_rule_distillation/2026-07-22-14-43-06_te_shape_first_distilled_periodic_mlp_harmonic_fw_polished_setpoints`